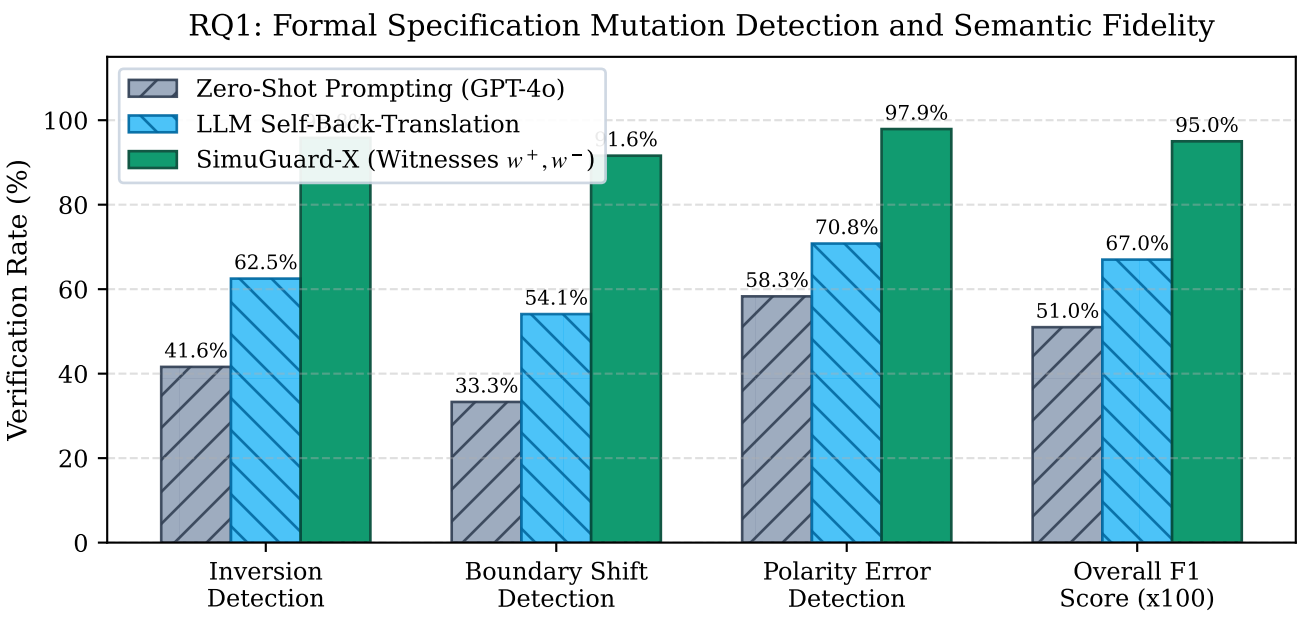
RQ1: Formal Specification Mutation Detection and Semantic Fidelity
Zero-Shot Prompting (GPT-4o)
LLM Self-Back-Translation
SimuGuard-X (Witnesses w^+, w^-)
Verification Rate (%)
0
20
40
60
80
100
41.6%
62.5%
91.6%
33.3%
54.1%
97.9%
58.3%
70.8%
95.0%
51.0%
67.0%
Inversion Detection
Boundary Shift Detection
Polarity Error Detection
Overall F1 Score (x100)